COMP3074

Mobile Application Development I

FINAL PROJECT
APPLICATION DESIGN PROPOSAL

Application Name: _____

Team ID (from D2L): _____

Submission Date: _____

Group size: maximum 5 students

Project Information

#	Student Name	Student ID	Primary Role / Responsibility
1	Abdirahman, Zuheib	101549351	
2	Howlader, Munir	101172332	
3	Morrison, Ben	101572409	

1. Project Overview

Provide a brief description of the proposed mobile application. Explain what the application will do, who will use it, what problem, need, or opportunity it addresses, and why a mobile application is appropriate. Recommended length: 150–250 words.

The Campus Study Group Finder is an Android app that allows George Brown College students create, discover, and join study sessions for their courses. Students can post a session with the course code, topic, date, time, location (on campus or online), and number of available seats. Other students can browse and filter sessions by course or date, join with a single tap, and track their upcoming sessions in one place.

The primary users are college students, especially those in shared programs, large courses, or have difficulty meeting others. Secondary users include peer tutors and student leaders who run regular review sessions and need an easy way to announce them and manage attendance.

Finding study partners can be informal and scattered. Students rely on class group chats, Discord, word of mouth, or posts on Teams/d2l that get lost quickly. Students who are new, commute, or study part-time are often left out entirely. There is no simple way to see how many people plan to attend a session, which makes planning a room or meeting unreliable.

The app provides students a single, dedicated place to organize study sessions around their actual courses. This makes collaborative learning more accessible and helps students connect with classmates they wouldn't otherwise meet. It can also support academic success by making it easier to prepare for exams and assignments in groups. As the app is built around course codes and campus locations, it could later be expanded to other colleges or add features such as reminders, in-session chat, and peer tutoring support.

2. Project Objectives

Identify 3–6 main objectives. Describe what the completed application is expected to accomplish.

1. Centralize study session discovery. The application will give students one place to find study sessions for their courses. Students will be able to browse upcoming sessions, search by course code or topic, and filter by date, so they no longer have to depend on scattered group chats or class announcements.
2. Make creating and joining sessions quick and simple. Students will be able to create a study session in a few steps by entering the course, topic, date, time, location, and maximum number of participants. Other students will be able to join or leave with a single tap, and the available seat count will update so organizers know how many people to expect. Help students get to their sessions using location services

3. The application will use Google Maps and the device's GPS to show where each session takes place, and provide directions. This is intended to help new students and commuters who are less familiar with the campus.
4. Keep students organized with a personal session view. A "*My Sessions*" screen will list every session a student has created or joined, so they can keep track of upcoming commitments in one place.
5. Students will sign in with an account, and their data will be stored securely. Only the creator of a session will be able to edit or delete it. The interface will use a consistent visual style across all screens, and the application will handle errors gracefully so it does not crash.

3. Target Users

Describe the primary user group, their needs, and a typical usage scenario.

Primary User Group are college students enrolled at George Brown College, particularly those in large, shared, or online courses where they may not know many classmates. This includes students new to the college, commuters who spend limited time on campus, part-time students, and international students who are still building their academic networks. Many of these students want to study with peers but have no reliable way to find out when and where study sessions are happening.

Secondary user group are peer tutors, student leaders, and students who reguarly organize review sessions also benefit from the application. They need a simple way to announce sessions, reach students in a specifice course, and know how many people plan to attend. Instructors may also find this app beneficial for class group creation.

User needs

- A single place to find study sessions organized by course
- A quick way to join or leave a session without back-and-forth messaging
- Clear information about when and where a session takes place, including directions on campus, or a remote link
- A way to keep track of the sessions they have committed to
- Confidence that session information is accurate and can only be changed by its organizer

Typical usage scenario

Aisha is a first-year student in a programming course with a midterm coming up next week. She doesn't know many classmates and wants to review with others. She opens the application, signs in, and searches the session feed for her course code. She finds a review session scheduled for Thursday afternoon in a campus study room, with three seats still open. She opens the session details, reads the topic description, and taps Join. The seat count updates, and the session is added to her My Sessions list. On Thursday, she opens the session from My Sessions, views the location on the map, and follows the direction sfrom her current location to the study room.

Later that week, Aisha decides to organize her own session for an upcoming assignment. She creates a new session by entering the course code, topic, date, time, location, and maximum number of participants. Classmates searching for that course can now find and join it, and Aisha can see how many people are attending.

4. Functional Requirements

List the major functions the application will provide. Use Must Have, Should Have, or Could Have for priority.

ID	Functional Requirement	Priority
FR-01	Users can create an account and sign in with their college email	Must Have
FR-02	Users can sign out, and the app remembers a sign-in user between launches	Must Have
FR-03	Users can create a study session with a course code, topic, date, time, location, and maximum number of participants	Must Have
FR-04	The app validates session input (required fields, future dates, seat limits above zero) before saving	Must Have
FR-05	Users can view a list of upcoming study sessions	Must Have
FR-06	Users can search sessions by course code or topic	Must Have
FR-07	Users can view full details of a session, including the current seat count	Must Have
FR-08	Users can join or leave a session, and the seat count updates accordingly	Must Have
FR-09	The app prevents users from joining a session that is full	Must Have
FR-10	Users can view a list of sessions they have created or joined (My Sessions)	Must Have
FR-11	Users can view a session's location on a map	Must Have
FR-12	Users can get directions from their current GPS location to a session	Must Have
FR-13	Only the creator of a session can edit or delete it	Must Have
FR-14	The app displays a splash screen and an About screen listing team members	Must Have
FR-15	Users can filter sessions by date	Should Have
FR-16	Users can view and edit their	Should Have

	profile (name, program, enrolled courses)	
FR-17	Users can share session details by email	Should Have
FR-18	Previously loaded sessions remain viewable when the device is offline	Should Have
FR-19	Users review a reminder notification before a joined session starts	Could Have
FR-20	Users can take or upload a profile photo using the camera	Could Have
FR-21	Users can add a joined session to their device calendar	Could Have
FR-22	Users can rate a session after attending	Could Have

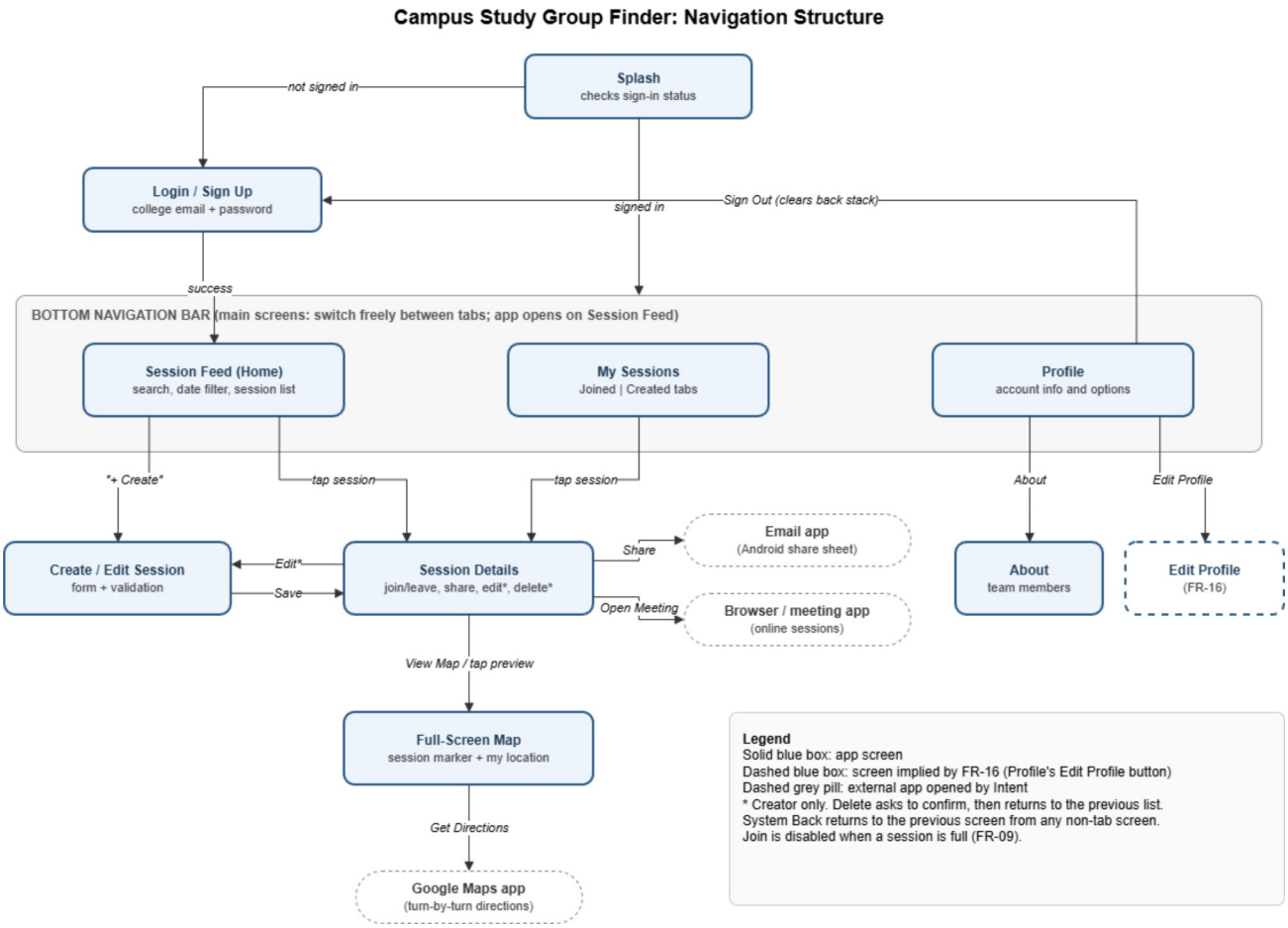
5. Application Screens and Navigation

Identify the main screens and describe how users navigate between them. Include a simple navigation diagram.

Screen	Purpose	Main UI Components
Splash	Displays the app logo while the app loads and checks whether the user is already signed in	App logo (ImageView), app name (TextView), loading indicator (ProgressBar)
Login / Sign Up	Lets users sign in with their college email or create a new account	Email and password fields (EditText), Login buttons, Sign Up button, toggle link between login and sign-up modes (TextView), error messages
Session Feed (Home)	Shows upcoming study sessions and lets users search for sessions by course or topic	Search bar (SearchView), session list (RecyclerView with session cards), Create Session button (FloatingActionButton), bottom navigation bar
Session Details	Shows full information about a selected session and lets users join or leave it	Course code, topic, description, date, time, location, and organizer (TextViews), seat count, map preview (MapView), Join/Leave button, Edit and Delete buttons (creator only), Share button
Create/Edit Session	Lets users create a new session or update one they created	Course code, topic, and description fields (EditText), date picker, time picker, location field, max participants field (number input), Save button, Cancel button,

		validation messages
Full-Screen Map	Shows the session location and the user's current position, with directions	Google Map (MapFragment), session location marker, current location marker, Get Directions button
My Sessions	Lists the sessions the user has joined or created	Tabs for Joined or Created (TabLayout), session list (RecyclerView), empty-state message, bottom navigation bar
Profile	Shows the user's account information and account options	Name, email, program, and enrolled courses (TextViews), Edit Profile button, About button, Sign Out button, bottom navigation bar
Edit Profile	Lets users update their profile details and the courses they're enrolled in, so the information shown on their Profile stays accurate	Name and program fields (OutlineTextField), read-only email (Text), course code field with Add button, enrolled course list as removable chips (InputChip), Save button, Cancel button, validation messages, top bar with Back
About	Displays app information and the names of all team members	App logo (ImageView), app name and version (TextView), team member names (TextViews)

Navigation diagram:



6. Wireframes / UI Mockups

Provide a wireframe or mockup for each major screen. Label important UI components and briefly explain what the user can do on each screen. Wireframes may be created using Figma, diagrams.net, another design tool, or clear hand-drawn sketches.

Screen 1: Splash

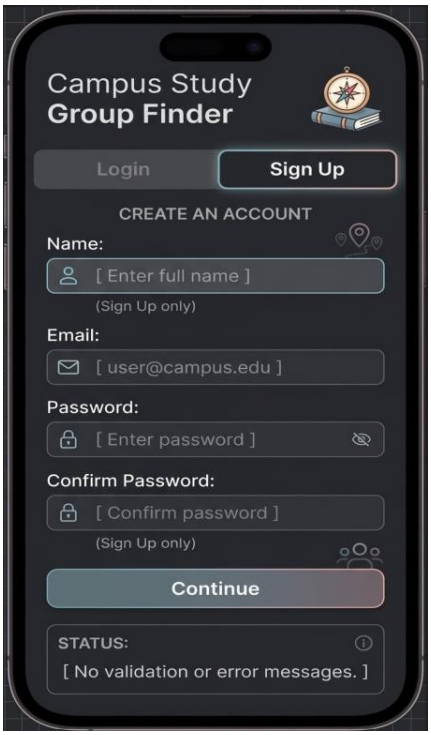
Purpose: Display the application identity while checking the user's sign-in status.



Logo and Name is to be finalized later.

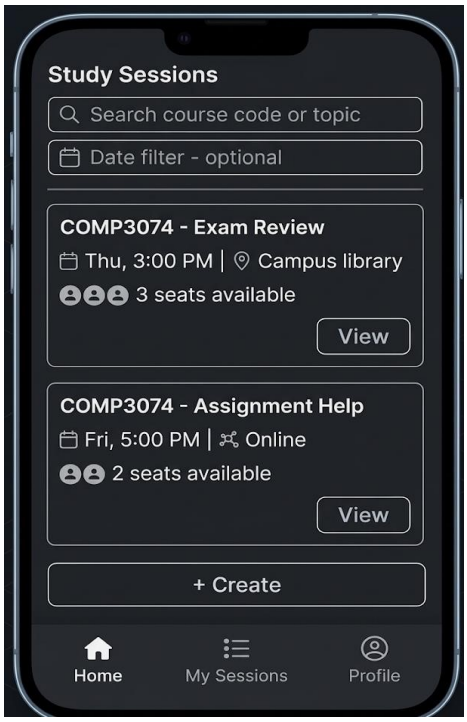
Screen 2: Login / Sign Up

Purpose: Allow students to sign in or create an account using their college email.



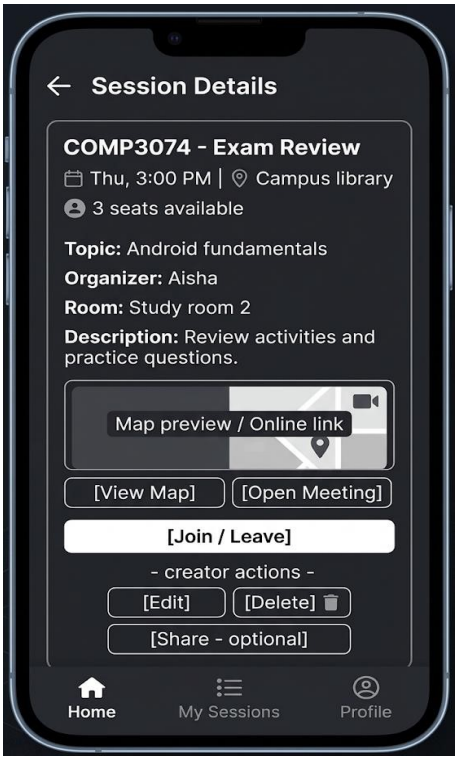
Screen 3: Session Feed

Purpose: Help students discover upcoming study sessions.



Screen 4: Session Details

Purpose: Purpose: Display session information and allow users to manage attendance.



7. Data Design

Describe the information the application needs to store or manage. Identify the main data entities/classes. Add additional tables as needed. If relationships exist, include a simple data model or class diagram.

Entity / Class Name: user

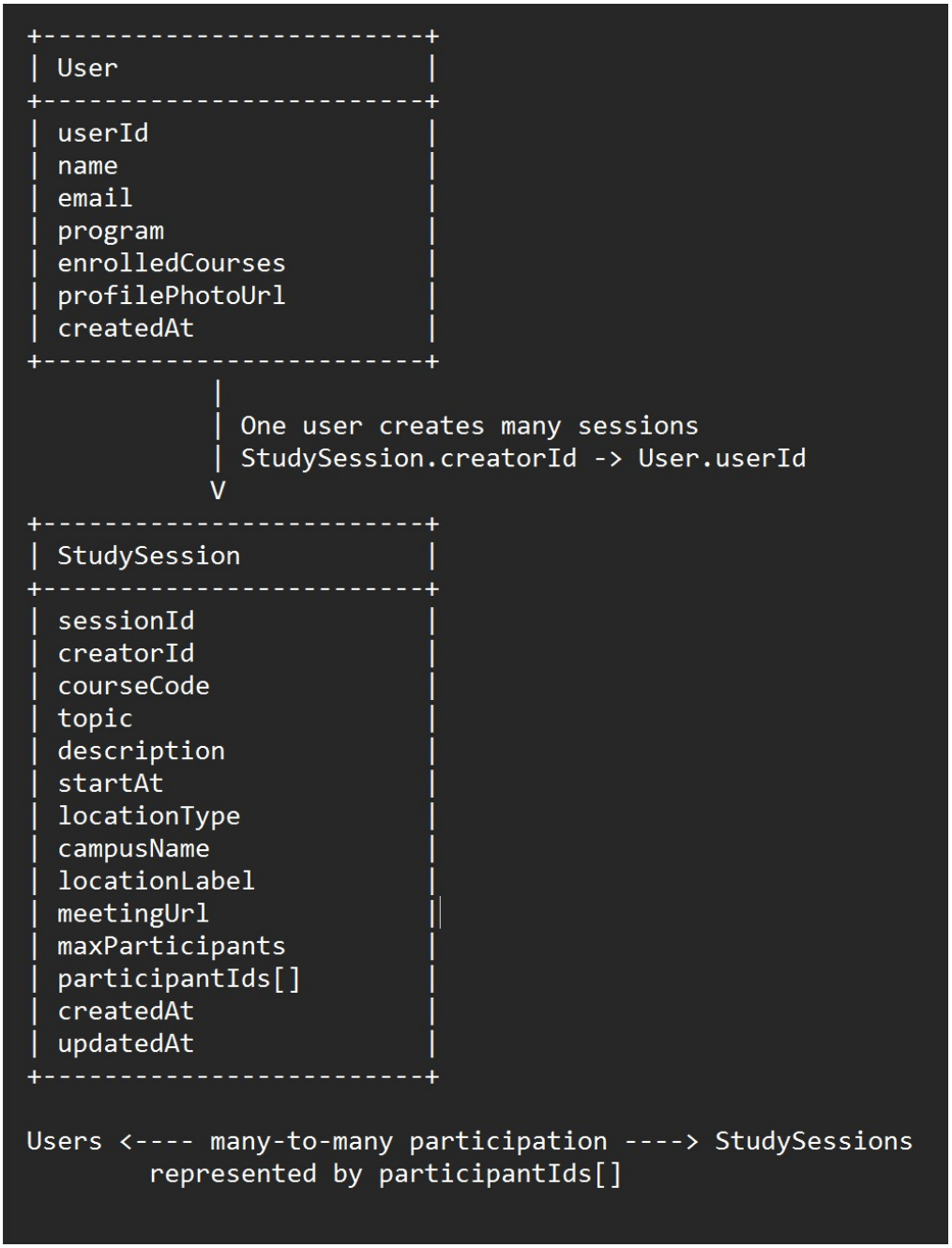
Attribute	Data type	Description
userId	String	Unique identifier
name	String	Display name entered during registration.
email	String	College email address associated with the account.
program	String	Academic program
enrolledCourses	String,	Course codes saved in the user's profile.
profilePhotoUrl	String	Reference to a profile image (if photo feature is implemented.)
createdAt	Timestamp	Date and time the profile was created.

Entity / Class Name: StudySession

Attribute	Data type	Description
sessionId	String	Unique session identifier; also used as the document ID.
creatorId	String	User ID of the organizer.
courseCode	String	Normalized course code, such as COMP3074.
topic	String	Short session title or study topic.
description	String	Additional information about the session.
startAt	Timestamp	Session start date and time, stored as a single timestamp.
locationType	String	Either ON_CAMPUS or ONLINE.
campusName	String	Campus name; required for an on-campus session.
locationLabel	String	Building, address, and room details for an on-campus session.
meetingUrl	String	Meeting link; required for an online session.

Attribute	Data type	Description
maxParticipants	Integer	Maximum total attendance, including the organizer.
participantIds	String	Unique user IDs of attendees, including the organizer.
createdAt	Timestamp	Date and time the session was created.
updatedAt	Timestamp	Date and time the session was last changed.

Data model / class diagram:



8. External APIs and Services

Identify external APIs, web services, AI services, or third-party libraries. Explain why each is needed, how the app communicates with it, what data is exchanged, and whether authentication/API keys are required. If none are planned, state this explicitly. Do not include API keys, passwords, or access tokens.

API / Service	Purpose	Data Sent	Data Received
Fire base authentication	Register and sign in users with their college emails, keep them signed in between launches and sign them out.	Email address, passwords (during sign ins) and the current ID token.	User ID, email adress,authentication tokens and success/ error responses
Fire base cloud firestore	Store and sync user profiles and study session data in real time so seat counts update for users instantly	User profile fields (Name,email, program enrolled in)	Real time session documents, user document and updated seat counts
Google maps SDK for Android (Maps Compose)	Display a sessions campus on a map inside the sessions details area.	Map camera position (latitude/longitude), marker coordinates, and restricted maps API Key.	Map tiles,map rendering data, and marker overlay data.

9. Technical Design

Summarize the proposed technical implementation.

Development Platform: Android Studio

Programming Language: Kotlin

Minimum Android Version / API Level: API 26 (Android 8.0)

Data Storage: Firebase Cloud Firestore (NoSQL database for real-time session updates) and Firebase Authentication for college email sign-ins.

Architecture / Application Structure: MVVM (Model-View-ViewModel) architecture. UI layouts will be constructed declaratively using Jetpack Compose.

External Libraries: Google Maps SDK for Android (via the Maps Compose library), Google Play Services Location API for GPS routing.

Briefly explain how the major parts of the application will work together: The application will authenticate users via Firebase using their college email credentials. The UI, built declaratively with Kotlin and Jetpack Compose in Android Studio, will observe state from ViewModels that retrieve real-time StudySession data from Firestore. When a user creates or joins a session, the app validates the input (checking future dates and available seats) and pushes the update to Firestore, which immediately syncs the updated seat count to all clients. The Google Maps SDK will be integrated directly into the Session Details and Full-Screen Map

composable functions to render location coordinates and calculate directions from the device's current GPS location.

10. Team Responsibilities

Describe the initial division of work. All members are expected to contribute to implementation and understand the overall application.

Team Member	Primary Responsibilities
Ben Morrison	Sections 1–5 (overview, objectives, target users, functional requirements, screens and navigation diagram). **Development (Architecture & Location)**: MVVM project structure, Compose navigation, app theme and styles, Splash and About screens, map preview, Full-Screen Map, location permissions and directions, My Sessions screen.
Munir Howlader	Sections 6, 7, 9, 13 and 14 (wireframes, data design, technical design, MVP, optional features). **Development (Data & Discovery)**: User and StudySession models, Firestore repositories and real-time listeners, security rules, Session Feed with search and date filter, Session Details, join/leave transactions and seat limits, share by email.
Zuheib Abdriahman	Sections 8, 10, 11, 12, 15 and 16 (APIs, team responsibilities, development plan, risks, references, declaration). **Development (Auth & Session Management)**: Firebase Authentication, Login/Sign Up, persistent sign-in and sign-out, Create/Edit Session with validation, creator-only edit/delete, Profile and Edit Profile screens.

11. Development Plan

Provide an initial development plan showing that the project can reasonably be completed within the available course timeframe.

Stage / Milestone	Planned Work	Responsible Member(s)	Target Date
Proposal and set up	Finalize the proposal, create the GitHub repository, and configure the android studio project, add the fire base project, enable authentication, and fire store and add	All members	Week 1

	the maps API key securley.		
Project skeleton	Set up the MVVM package structure, Jetpack Compose navigation graph, app theme, and placeholder screens for Splash, Login/Sign Up, Session Feed, Session Details, My Sessions, Profile, and About.	Ben Morrison	Week 2
Authentication	Implement the Splash sign-in check, Login and Sign Up screens, Firebase Authentication integration, persistent sign-in, sign-out.	Zuheib Abdirahman	Week 3
Data Layer	Define the User and StudySession data classes, build Firestore repositories, and connect real-time listeners for live data updates.	Munir Howlader	Week 4
Session Creation and Management	Build the Create and Edit Session screens, validate required fields, future dates and seat limits, and enforce creator-only edit and delete rules.	Zuheib Abdirahman	Week 5
Session Discovery	Implement the Session Feed, search by course code or topic, and the optional date filter.	Munir Howlader	Week 6

12. Risks and Challenges

Identify at least three potential challenges and explain their impact and mitigation.

Risk / Challenge	Potential Impact	Proposed Solution / Mitigation
Real-time seat-count conflicts (race conditions). Two users try to join the last available seat at the same time, causing overbooking or an inconsistent	High. This breaks FR-09, damages user trust, and creates a poor experience where more people show up than the room can hold	Use Firestore transactions for all join and leave operations so the seat count is read and updated atomically. Add server-side security rules that reject

seat count.		any write exceeding maxParticipants. Test with simulated concurrent joins before the final demo.
Firebase setup and security rules misconfiguration. Incorrect Firestore rules could expose user data or block legitimate writes, and API keys could accidentally be leaked.	High. Creates a security vulnerability, risks data loss, or causes app-breaking permission errors that are difficult to debug late in the project.	Define Firestore security rules early (only the creator can edit or delete a session; only authenticated users can read sessions). Store API keys in local.properties and BuildConfig, restrict the Maps key to the app's package name and SHA-1, and never commit secrets to GitHub. Use the Firebase emulator suite for testing.
Location permissions and GPS or map failures. Users deny location permission, GPS is unavailable indoors, or the Maps API key fails to load map tiles.	Medium. FR-11 and FR-12 become unusable, and users cannot find or navigate to campus sessions.	Request permissions at the point of use with a clear rationale. Provide graceful fallbacks such as showing the text address and an "Open in Google Maps" link if permission is denied. Cache the last known location and handle null results. Test on both emulator and physical devices.

13. Minimum Viable Product (MVP)

Define the minimum version of the application that the team considers a successful implementation. List the features that must work.

1. Account registration and authentication. Students can create an account using an approved college email address, sign in, remain signed in between launches, and sign out. Invalid inputs and authentication errors produce clear messages. Covers FR-01 and FR-02.

2. Session creation and organizer management. Students can create a campus or online session with a course code, topic, future date and time, location or meeting link, and participant limit. Only the creator can edit or delete it, and required fields and attendance limits are validated. Covers FR-03, FR-04, and FR-13.

3. Session discovery and details. Students can browse upcoming sessions, search by course code or topic, and view full session information, including available seats. Covers FR-05, FR-06, and FR-07.

4. Participation and personal session tracking. Students can join and leave sessions without duplicate registrations or exceeding capacity. My Sessions displays sessions they have joined or

created, and attendance changes are reflected in the session details. Covers FR-08, FR-09, and FR-10.

5. Location support and required application screens. Campus sessions display a location map and provide access to directions from the user's current location. Online sessions provide a usable meeting link instead. The application includes a splash screen, an About screen listing all team members, and consistent navigation between its main screens. Covers FR-11, FR-12, and FR-14.

14. Optional / Future Features

List features the team would like to implement if sufficient development time remains. These should not be required for the basic operation of the application.

1. Push notification reminders sent to users before a joined session begins.
2. Profile photo integration allowing users to capture or upload an image using the device camera.
3. Direct calendar integration to add joined sessions into the device's native calendar.
4. A post-session rating system allowing attendees to review the quality of the study group.
5. AI note taking integration, where AI will listen and take notes of all the study discussion.

15. References and Resources

List Android/API documentation, libraries, tutorials, articles, design resources, and AI tools used in preparing the proposal. Use a consistent citation format.

Google. "Android Developers Build your first app." Android Developers. https://developer.android.com/training/basics/firstapp (accessed [date]).

Google. "Jetpack Compose Android Developers." https://developer.android.com/jetpack/compose (10/02/26).

Google. "Guide to app architecture — Android Developers." https://developer.android.com/topic/architecture (10/02/26).

Google. "Firebase Authentication — Firebase Documentation." https://firebase.google.com/docs/auth (10/02/26).

Google. "Cloud Firestore Firebase Documentation." https://firebase.google.com/docs/firestore (10/02/26).

Google. "Maps Compose Library — GitHub." https://github.com/googlemaps/android-maps-compose (10/02/26).

Google. "Google Maps SDK for Android — Maps Platform Documentation." https://developers.google.com/maps/documentation/android-sdk (10/02/26).

Google. "Location API — Android Developers." https://developer.android.com/develop/sensors-and-location/location 10/02/26

16. Team Declaration

1. _____

2. _____

3. _____

4. _____

5. _____

Submission Checklist

☐ Project title and team information

☐ Project overview and objectives

☐ Target users

☐ Functional requirements

☐ Screen/navigation design

☐ Wireframes

☐ Data design

☐ API/service information, if applicable

☐ Technical design

☐ Team responsibilities

☐ Development plan

☐ Risks and challenges

☐ MVP definition

☐ Optional features

☐ References